

Slight Homomorphic Signature for Access Controlling in Cloud Computing

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Abstract With the popularity of cloud computing, how to securely authenticate a user while not releasing user's sensitive information becomes a challenge. In this paper, we introduce a slight homomorphic signature, which is suitable to implement an access controlling service in cloud computing. In slight homomorphic signature, each user in cloud computing who have a set of identity attributes, firstly computes a full signature on all his identity attributes, and sends it to a semi-trusted access controlling server. The access controlling server verifies the full signature for all identity attributes. After then, if the user wants to require a cloud service, which may have a special requirement on one of the identity attributes, the user only needs to securely send the cloud service's name to the access controlling server. The access controlling server which does not know the secret key can compute a partial signature on this special identity attribute, and then sends it to the cloud server for authentication. In the paper, we give a formal secure definition of this slight homomorphic signature, and construct a scheme from Boneh–Boyen signature. We prove that our scheme is secure under q -SDH problem with a weak adversary.

Keywords Slight homomorphic signature · Identity authentication · Cloud computing · Boneh–Boyen signature · q -SDH assumption

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1 Introduction

With the popularity of cloud computing, many secure problems have been identified. One of the most important problems is that it is a big challenge to authenticate a cloud user to a cloud service without releasing user's sensitive information to cloud service providers when it is unnecessary. In previous work, a trusted third party has been used as an access control server to help the cloud user. In this scheme, the user has to share a lot of information with the trusted third party, which is very risky in cloud computing. Other efforts have been made to design identity manager to protect user's sensitive information, while most approaches are still using trusted third party and add a lot of overhead to the system thus degrades the performance of the system. Moreover, the cryptography researchers have done many work in this area. In 2009, Gentry proposed a fully homomorphic encryption scheme to make it possible to analyze data without decrypting it [12]. Those efforts make it possible to preserve user's sensitive information when the user gets authenticated. However, up to now, some homomorphic encryption schemes have been proposed [12,20,21], while only a few homomorphic signature schemes have been presented.

In General, a homomorphic signature scheme [15] includes a set of private keys $k_1, k_2, \dots, k_n$, a set of public keys $k'_1, k'_2, \dots, k'_n$, and a signature algorithm and a verification algorithm. In this cryptographic primitive, given a set of signature $\sigma_1, \dots, \sigma_n$ on messages $m_1, \dots, m_n$ in message space $\mathcal{M}$, a function $f : \mathcal{M}^n \rightarrow \mathcal{M}$ from a set of "admissible" functions $\mathcal{F}$, one can compute a signature on the pair $(f, f(m_1, \dots, m_n)) \in \mathcal{F} \times \mathcal{M}$. If the signature is valid, then it means that the claimed value equals $f(m_1, \dots, m_n)$. Homomorphic signature can be used to cloud computing, especially in implementing an access controlling service.

In this paper, our goal is to design a novel homomorphic signature scheme, slight homomorphic signature, suitable for implementing access controlling in cloud computing. One of our motivations is to authenticate a cloud user to a cloud service without disclosing user's sensitive information. The other motivation is to reduce the computation overhead at the client side in the procedure of the security protection, because most clients may not be so powerful machines such as tablets and pads. In cloud computing, each user usually has a set of identity attributes, and to get authenticated to a cloud service, the user needs to provide a part of his information for identify verification based on the service's requirements. For example, in Teamviewer,¹ if a cloud service user needs to take control of a family member's computer, he can only authenticate his family role to the Teamviewer based on the provided necessary identity attributes, without disclosing his other identity information.

In our scheme, instead of using a trusted third party, the user firstly computes a full signature on all his identity attributes, and sends it to a semi-trusted access controlling server. The access controlling server verifies the full signature for all identity attributes. When the user wants a cloud service, the user securely sends the cloud service's name to the access controlling server. The access controlling server which does not know the secret key can compute a partial signature on this special identity attribute, and send it to the cloud server for authentication. Service will be granted if the cloud server verifies the provided partial signature and the user does not release all his identity attributes.

The contributions of the paper can be listed as follows: 1) We give a security definition of the slight homomorphic signature; 2) We construct such a slight homomorphic signature

¹ Teamviewer is one of the top 15 consumer cloud services, which is a great screen sharing and "GoToMyPC" style service. The service works PC to PC, Mac to Mac, and PC to Mac as well. If used in a commercial setting, license fees are a one-time charge and range from 7,500 to 2,700 for corporate accounts.

scheme based on the Boneh–Boyen signature [4] ; 3) We provide a security proof to our scheme with a weak adversary, using q -SDH assumption.

The rest of the paper is organized as follows. Section 2 shows the application of the slight homomorphic signature, which is followed by Sect. 3 describing the related work of homomorphic signature. Section 4 reviews a list of related concepts with our scheme and Sect. 5 gives the definition of slight homomorphic signature. The construction of the proposed scheme is discussed in Sect. 6. Section 7 shows the security of our scheme and finally Sect. 8 concludes the paper.

2 Application of Slight Homomorphic Signature

In cloud computing, users usually have multiple identities associated with a single or multiple cloud service providers. Usually, those identity attributes are provided when the users are registered. To use a cloud service, a user needs to authenticate herself to it. The user has to provide some identity attributes (e.g., name, home address, credit card number, phone number, driver's license number, date of birth, etc.) according to the requirements of the specific cloud service, when the user is requesting cloud service from the service providers. Obtaining the user's identity attributes helps the cloud service provider to decide whether to grant access to its service or not. However, if all identity attributes are disclosed, as a consequence those information can be used to uniquely identify, contact, or locate a particular user, which, if not properly protected, may be exploited and abused. Thus, when requesting a cloud service, a user does not want to disclose all his identity attributes, but only provides a part of this information according to the requirements of the cloud service.

Different solutions use different ways of sending user's identity attributes for negotiation with the cloud service providers. The common way is to use a trusted third party (Access controlling server) as shown in Fig. 1. The trusted third party should know user's secret key, so that it can authenticate the user for cloud service providers [18]. However, in the real world, fully trusted third party may not exist. Recent security breach at Dropbox shows

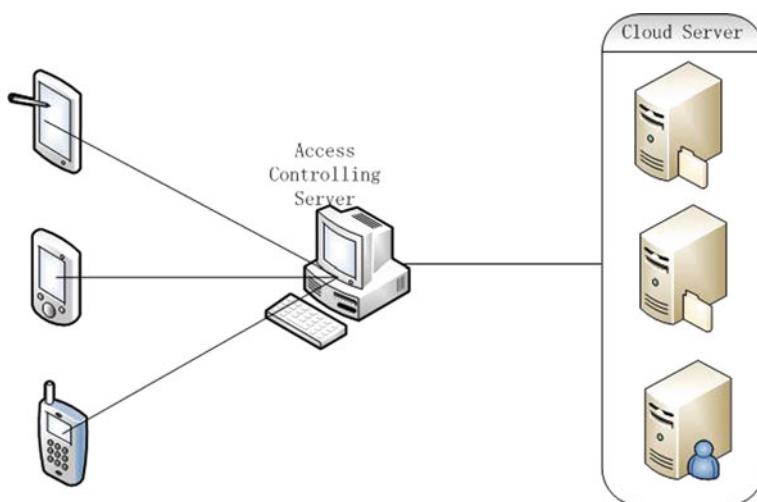


Fig. 1 Access controlling in cloud computing

the difficulty in building a trusted third party. In this paper, we try to use the homomorphic cryptography to solve this problem.

In our slight homomorphic signature, we assume that every user has a set of identity attributes $\mathbf{v} = \{v_1, \dots, v_q\}$. Firstly, user pre-computes the full signature σ based on all his identity attributes $(v_1, \dots, v_q)$, and sends them to the access controlling server, which is a partially trusted third party. Later on, when the user wants to require a cloud service, which has a special requirement on v_i , the user will get the authentication with the help of the access controlling server. For this purpose, the cloud user needs to securely send the cloud service's name to the access controlling server. Then, with stored full signature and cloud service's name, the access controlling server can compute a partial signature $\hat{\sigma}$ on the v_i by using the slight homomorphic signature algorithm, after then it sends $(\hat{\sigma}, v_i)$ to cloud server for authentication. The cloud server verifies the partial signature using the verification algorithm and grants the service if the verification is successful.

There are two advantages for the above application of slight homomorphic signature. Firstly, the semi-trusted access controlling sever does not know the secret key, which can protect user's privacy. Secondly, during the access controlling process, the user does not need to do any heavy computation, and only secretly sends name of cloud service to the access controlling sever. The pre-computation of full signature can be carried with the help of other servers.² Thus, it is very suitable for the current cloud computing where a lot of users are devices with limited resource, such as smartphones, pads and tablets.

3 Related Work

Homomorphic signature was firstly proposed by Johnson et al. [15] In the past few years, there are about three classes research have touch on this area:

Quoting/redacting signature [1,9,10,13,16] is that given Alice's signature on some message m , any one can derive Alice's signature on a substring of m . Quoting signature can be applied to signed text messages where one wants to derive Alice's signature on a substring of m . Quoting can also apply to signed images where one wants to derive a signature on a subregion of the image and to data structures where one wants to derive a signature of a subset of the data structure such as a sub-tree of a tree.

Arithmetic signature [5–7,11,22] is motivated by the application of secure network coding. Given Alice's signature on vectors $v_1, \dots, v_k \in \mathbb{F}_p^n$, one should be able to derive Alice's signature on a vector $\mathbf{v}$ in the linear span of $v_1, \dots, v_k$. Recent constructions go beyond linear operations and support low degree polynomial computations [6].

In **Transitive signature** [2,14,17,19,23], given Alice's signature on edges in a graph G , any one can derive Alice's signature on a pair of vertices u, v , if there exists a path from u to v in G . The derived signature on the pair (u, v) must be indistinguishable from a fresh signature on (u, v) had Alice generated one herself.

In this paper, we put forth a notion of **slight homomorphic signature**, which is a special case of arithmetic signature. Using slight homomorphic signature, given Alice's full signature σ on her all identity attributes $v_1, \dots, v_n$, anyone can generate the partial signature σ_i on the special identity attribute v_i .

Digital Identity Manager [3,18], several efforts have been made to improve digital identity manager for cloud computing. Most of them focus on designing a trusted third party.

² Since the client side usually has limited resource, the heavy operations in the computation of full signatures can be forwarded to other servers.

Bertino et al. [3] proposes zero-knowledge proof protocols and semantic matching techniques to design a digital identity manager. Ranchal et al. [18] uses a predicate encryption scheme and multi-party computing to manage digital identity. Both proposed approaches are more complicated and have more overhead than our proposed scheme.

4 Preliminaries

4.1 q-SDH Assumptions

Firstly, we briefly review the necessary facts about bilinear maps and groups, in the notation of [8]. A tuple $\mathcal{G} = (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, \hat{e})$ is a bilinear group, if it has the following properties:

1. p is a prime;
2. $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ are three cyclic groups of prime order p ;
3. g_1 is a generator of $\mathbb{G}_1$ and g_2 is a generator of $\mathbb{G}_2$;
4. $\hat{e} : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ is a bilinear pairing i.e., a map satisfying the following properties:
 - (a) Bilinearity: $\forall u \in \mathbb{G}_1, \forall v \in \mathbb{G}_2, \forall a, b \in \mathbb{Z}, \hat{e}(u^a, v^b) = \hat{e}(u, v)^{ab}$;
 - (b) Non-degeneracy: $\hat{e}(g_1, g_2) \neq 1$ and is thus a generator of $\mathbb{G}_T$.

All group operations and the bilinear map must be efficiently computable. Formally, one defines a bilinear group generation algorithm that takes as input a security parameter $\lambda \in \mathbb{Z}^+$ and outputs the description of groups $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ and a bilinear map $\hat{e} : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$. We then require the existence of probabilistic polynomial time algorithms for computing the group operation in $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ and the bilinear map $\hat{e}$.

Secondly, we define the q-SDH problem in bilinear groups:

- **q-SDH Problem:** an instance of the q-SDH problem for $q \geq 1$ is a tuple $(g_1, g_1^\alpha, g_1^{\alpha^2}, \dots, g_1^{\alpha^q}, g_2, g_2^\alpha)$ for randomly choose generators $g_1 \xrightarrow{R} \mathbb{G}_1, g_2 \xrightarrow{R} \mathbb{G}_2$, and random number $\alpha \xrightarrow{R} \mathbb{Z}_p$. A solution is a pair $(r, g_1^{1/(\alpha+r)}) \in \mathbb{Z}_p \times \mathbb{G}_1$.

If $\mathcal{A}$ is an algorithm which can solve the q-SDH problem, we define $q\text{-SDH-ADV}_{\mathcal{A}}$ to be the probability that $\mathcal{A}$ outputs a solution. We say the q-SDH assumption holds if for all polynomial-time algorithm $\mathcal{A}$, $q\text{-SDH-ADV}_{\mathcal{A}}$ is negligible.

4.2 Boneh–Boyen Signature

Since our scheme is based on the signature proposed by Boneh and Boyen [4], we will review the signature in this section. Let $\mathcal{G} = (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, \hat{e})$ be a bilinear group. The Boneh–Boyen signature can be depicted as follows:

Setup(1^λ) This algorithm chooses random number $\alpha \xrightarrow{R} \mathbb{Z}_p$ and generators $g_1 \xrightarrow{R} \mathbb{G}_1, g_2 \xrightarrow{R} \mathbb{G}_2$. The public key is $pk = (\mathcal{G}, g_1, g_2, g_2^\alpha)$, and the secret key $sk = \alpha$.

Sign(sk, m) This algorithm outputs signature $\sigma = g_1^{1/(\alpha+m)}$.

Verify(pk, m, σ) This algorithm outputs 1 if $\hat{e}(\sigma, g_2^m) \cdot \hat{e}(\sigma, g_2^\alpha) = \hat{e}(g_1, g_2)$, otherwise outputs 0.

Boneh–Boyen signature is proved secure in the standard model, but its security is under a stronger assumption, the q-SDH assumption. The above version is secure against weak adversary, and we denote it BB.

Theorem 4.1 *If the q -SDH assumption holds, then BB is unforgeable against a weak adversary making at most q signature queries.*

5 Slight Homomorphic Signature for Access Controlling in Cloud Computing

We denote the slight homomorphic signature for access controlling in cloud computing as Σ signature. In a Σ signature scheme, user computes full signature σ on his all identity attributes $(v_1, \dots, v_n)$. The access controlling server should verify σ for all $(v_1, \dots, v_n)$. Then, according to the instruction from user, the access controlling server can generate the partial signature σ_i on the special identity attribute v_i . Formally, we have the following:

Definition 5.1 A Σ signature scheme is a tuple of probabilistic, polynomial-time algorithms (**Setup**, **Sign**, **Eval**, **Verify1**, **Verify2**) as follows:

- **Setup** $(1^\lambda, q)$: Takes as input a security parameter λ and a maximum identity attributes set size q (This q is the same as q signature queries in Sect. 4.2.), outputs a public key pk and a secret key sk .
- **Sign** $(sk, \{v_1, \dots, v_q\})$: Takes a secret key sk , a set of identity attributes $\{v_1, \dots, v_q\}$, outputs a signature σ .
- **Eval** (pk, v_i, σ) : Takes a public key pk , an identity attribute v_i , and a full signature σ , verifies σ for $\{v_1, \dots, v_q\}$, and then outputs a partial signature σ_i on v_i .
- **Verify1** $(pk, \{v_1, \dots, v_q\}, \sigma)$: Takes a public key pk , a set of identity attributes $\{v_1, \dots, v_q\}$, and a full signature σ , outputs either 0(reject) or 1(accept).
- **Verify2** (pk, v_i, σ_i) : Takes a public key pk , an identity attribute v_i , and a partial signature σ_i , outputs either 0(reject) or 1(accept).

Informally, the correctness of our scheme are that (1) a full signature σ generated by **Sign** verifies for $\{v_1, \dots, v_q\}$, and (2) a partial signature σ_i output by **Eval** on verifies for v_i . Formally, we require that for each (pk, sk) output by **Setup** $(1^\lambda, q)$, we have:

1. If $\sigma \leftarrow \text{Sign}(sk, \{v_1, \dots, v_q\})$, then with overwhelming probability $\text{Verify1}(pk, \{v_1, \dots, v_q\}, \sigma) = 1$.
2. For all $v_i \in \{v_1, \dots, v_n\}$, if $\sigma_i \leftarrow \text{Eval}(pk, v_i, \sigma)$, then with overwhelming probability $\text{Verify2}(pk, v_i, \sigma_i) = 1$.

5.1 Security

The goal of an adversary attacking a Σ signature scheme is to forge a valid signature on an identity attribute $v^* \notin \{v_1, \dots, v_q\}$. In our security model, the adversary should query $\{v_1, \dots, v_q\}$ all in advance (weak adversary), and the challenger will give the response. After querying, if the adversary can finally provide a valid signature on $v^* \notin \{v_1, \dots, v_q\}$, then the challenger can solves the hard (q -SDH) problem (Since it is impossible, we can prove our scheme is secure).

Definition 5.2 A Σ signature scheme $\mathcal{S} = (\text{Setup}, \text{Sign}, \text{Eval}, \text{Verify1}, \text{Verify2})$ is unforgeable if for all q the advantage of any probabilistic, polynomial-time, weak adversary $\mathcal{A}$ in the following game is negligible in the security parameter q :

Setup: The challenger runs $\text{Setup}(1^\lambda, n)$ to obtain (pk, sk) and sends pk to $\mathcal{A}$.

Queries: $\mathcal{A}$ submits at most q identity attributes $v_i \in \{v_1, \dots, v_n\}$ for querying. For v_i , the challenger gives to $\mathcal{A}$ the signature $\sigma_i \leftarrow \text{Eval}(\cdot)$. This interaction is subject to the restriction that at most q identities can be queried.

Output: $\mathcal{A}$ outputs the forgery including a signature σ^* on an identity attribute v^* .

We say that the adversary wins if $\text{Verify2}(pk, v^*, \sigma^*) = 1$ and $v^* \notin \{v_1, \dots, v_n\}$. The advantage $\Sigma - \text{Adv}$ of $\mathcal{A}$ is the probability that $\mathcal{A}$ wins the above game. If $\Sigma - \text{Adv}$ is negligible, then the Σ signature scheme is secure.

6 A Σ Signature Scheme

We define a Σ signature scheme $\mathcal{S}$ from Boneh–Boyen signature scheme as follows:

S.Setup($1^\lambda, n$): On input a security parameter λ and a maximum identity attributes set size q , let $\mathcal{G} = (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, \hat{e})$ be a bilinear group, do the following:

1. Chooses random $\alpha \xrightarrow{R} \mathbb{Z}_p$ and generators $g_1 \xrightarrow{R} \mathbb{G}_1, g_2 \xrightarrow{R} \mathbb{G}_2$.
2. Output the public key $pk = (\mathcal{G}, g_1, g_2, g_2^\alpha)$ and secret key $sk = \alpha$.

S.Sign($sk, \{v_1, \dots, v_q\}$): On input a secret key sk and a set of identity attributes $\{v_1, \dots, v_q\}$, output

$$\sigma = g_1^{\frac{1}{(\alpha+v_1)\dots(\alpha+v_q)}},$$

and $\sigma^\alpha, \sigma^{\alpha^2}, \dots, \sigma^{\alpha^q}$.

S.Eval($pk, v_i, \{\sigma, \sigma^\alpha, \sigma^{\alpha^2}, \dots, \sigma^{\alpha^q}\}$): On input the public key pk , an identity attribute v_i , and the full signature $\sigma, \sigma^\alpha, \sigma^{\alpha^2}, \dots, \sigma^{\alpha^q}$, do the following:

1. Let $P_i(t) = \prod_{j=1, j \neq i}^q (t + v_j)$. Since $P_i(t)$ has the degree at most $q-1$, $\sigma_i = \sigma^{P_i(\alpha)}$ can be computed by using $\sigma, \sigma^\alpha, \sigma^{\alpha^2}, \dots, \sigma^{\alpha^q}$.
2. Compute the partial signature on v_i as $\sigma_i = \sigma^{P_i(\alpha)}$. Obviously, $\sigma_i = g_1^{1/(\alpha+v_i)}$.
3. Output σ_i .

S.Verify1($pk, \{v_1, \dots, v_q\}, \{\sigma, \sigma^\alpha, \sigma^{\alpha^2}, \dots, \sigma^{\alpha^q}\}$): Let $P(t) = \prod_{i=1}^q (t + v_i)$. Since $P(t)$ has the degree at most q , $\sigma^{P(\alpha)}$ can be computed by using $\sigma, \sigma^\alpha, \sigma^{\alpha^2}, \dots, \sigma^{\alpha^q}$. On input the public key pk , a set of identity attributes $\{v_1, \dots, v_q\}$, and the full signature $\{\sigma, \sigma^\alpha, \sigma^{\alpha^2}, \dots, \sigma^{\alpha^q}\}$, output 1 if $\sigma^{P(\alpha)} = g_1$; otherwise output 0.

S.Verify2(pk, v_i, σ_i): On input the public key pk , an identity attribute v_i , and a partial signature σ_i , output 1 if $\hat{e}(\sigma_i, g_2^{v_i}) \cdot \hat{e}(\sigma_i, g_2^\alpha) = \hat{e}(g_1, g_2)$; otherwise output 0.

Lemma 6.1 *The Σ signature scheme in Sect. 6 satisfies the correctness properties of Definition 5.1.*

Proof Suppose $\sigma \leftarrow \text{Sign}(sk, \{v_1, \dots, v_q\})$. Since $\sigma \leftarrow g_1^{\frac{1}{(\alpha+v_1)\dots(\alpha+v_q)}}$ and $P(\alpha) = \prod_{i=1}^q (\alpha + v_i)$, then **Verify1**($pk, \{v_1, \dots, v_q\}, \{\sigma, \sigma^\alpha, \sigma^{\alpha^2}, \dots, \sigma^{\alpha^q}\}$) = 1. Next, for all $v_i \in \{v_1, \dots, v_n\}$, suppose $\sigma_i \leftarrow \text{Eval}(pk, v_i, \{\sigma, \sigma^\alpha, \sigma^{\alpha^2}, \dots, \sigma^{\alpha^q}\})$, since $P_i(t) = \prod_{j \neq i}^q (t + v_j)$ and $\sigma_i \leftarrow \sigma^{P_i(\alpha)}$, which implies $\sigma_i = g_1^{1/(\alpha+v_i)}$, then

$$\hat{e}(g_1^{1/(\alpha+v_i)}, g_2^{v_i}) \cdot \hat{e}(g_1^{1/(\alpha+v_i)}, g_2^\alpha) = \hat{e}(g_1, g_2).$$

Thus, **Verify**(pk', v_i, σ_i) = 1.

Efficiency analysis: Many previous efforts study the issues of homomorphic signatures for restricted homomorphisms, and almost exclusively for linear functions, including [5–7, 11]. Up to now, the problem of fully homomorphic signatures under standard assumptions still remains open. However, in our application scenario such as in Teamviewer, it is required that anyone who has the full signature of Alice on her all identity attributes can generate a partial signature on a set of special identity attributes for Alice for the authentication purpose. Therefore, the general homomorphic signatures are not necessary and our proposed new definition of *slight* homomorphic signature is very suitable for this scenario. In our scheme, the computational cost of **Sign** algorithm is heavy, which needs q exponentiations over $\mathbb{G}_1$, but fortunately, the output value can be used repeatedly for a period time. Thus, this computational cost of **Sign** algorithm can be mostly ignored. Moreover, in our scheme, the most important algorithm is **Eval**, since it should be used very frequently. Fortunately, the **Eval** algorithm in our scheme only needs several multiplications by using $\{\sigma, \sigma^\alpha, \sigma^{\alpha^2}, \dots, \sigma^{\alpha^q}\}$. Compared with Boneh [6]’s scheme, which needs several exponentiations, our scheme is much more efficient in terms of computational complexity. $\square$

7 Security Analysis

In Definition 3.2, an adversary can break a Σ signature scheme by forging a valid signature (v^*, σ^*) , while $v^* \notin \{v_1, \dots, v_n\}$.

Theorem 7.1 *If q -SDH assumption holds, then our Σ signature scheme is weakly unforgeable. Specially, let $\mathcal{A}$ be an weak adversary that attacks our Σ signature scheme and outputs a forgery. Then there exists a challenger $\mathcal{C}$ that solves the q -SDH problem such that*

$$(1 - 1/p)\Sigma - \text{Adv}[\mathcal{A}] \leq q - \text{SDH} - \text{Adv}[\mathcal{C}].$$

Proof Our algorithm $\mathcal{C}$ will play the role of the attacker to q -SDH assumption and that of the challenger in the unforgeability game for our Σ signature scheme. Let tuple $\mathcal{G} = (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, \hat{e})$ be a bilinear group. Given $(g_1, g_1^\alpha, g_1^{\alpha^2}, \dots, g_1^{\alpha^q}, g_2, g_2^\alpha)$ for randomly choosing $g_1 \xrightarrow{R} \mathbb{G}_1, g_2 \xrightarrow{R} \mathbb{G}_2$, and $\alpha \xrightarrow{R} \mathbb{Z}_p$, $\mathcal{C}$ ’s goal is to output a pair $(r, g_1^{1/(\alpha+r)}) \in \mathbb{Z}_p \times \mathbb{G}_1$. When $\mathcal{A}$ submits $v_1, \dots, v_q$ for querying, $\mathcal{C}$ works as follows:

Setup:

1. Given distinct identity attributes $v_1, \dots, v_q$ queried by adversary, form the polynomial $P(t) = \prod_{i=1}^q (t + v_i) \in \mathbb{Z}_p[t]$.
2. Compute $x = g_1^{P(\alpha)}$ by using $g_1, g_1^\alpha, g_1^{\alpha^2}, \dots, g_1^{\alpha^q}$.
3. Send $\mathcal{A}$ the public key $pk = (\mathcal{G}, x, g_2, g_2^\alpha)$; the (implicit) secret key is α .

Queries: For each identity attributes $v_i \in \{v_1, \dots, v_n\}$, do the following:

1. Let $P_i(t) = \prod_{j \neq i} (t + v_j)$.
2. Compute the signature $\sigma_i = g_1^{P_i(\alpha)} = x^{1/(\alpha+v_i)}$ by using $g_1, g_1^\alpha, g_1^{\alpha^2}, \dots, g_1^{\alpha^q}$.
3. Send the signature σ_i to $\mathcal{A}$.

Forgery: When $\mathcal{A}$ outputs a forgery σ^* on $v^* \notin \{v_1, \dots, v_n\}$, $\mathcal{C}$ can also output a solution to q -SDH problem.

We analyze the simulation using a series of games. Let W_i be the event that $\mathcal{A}$ wins the unforgeability game in Game i .

Game 0. This is the real unforgeability game, where $\mathcal{A}$ interacts with a challenger $\mathcal{C}$ for our Σ signature scheme, and therefore $Pr[W_0] = \Sigma - Adv[\mathcal{A}]$.

Game 1. To compute a signature on v_i , we compute the σ_i as $\sigma_i = g_1^{P_i(\alpha)}$. It follows that

$$\sigma_i = x^{1/(\alpha+v_i)} = g_1^{P_i(\alpha)}.$$

Obviously, this change is only conceptual. Thus $Pr[W_1] = Pr[W_0]$.

Game 2. This is the same as Game 1, except that $\mathcal{A}$ outputs a forgery σ^* on v^* , where $v^* \notin \{v_1, \dots, v_n\}$. If the forgery is valid, then

$$\sigma^* = x^{1/(\alpha+v^*)} = g_1^{\frac{P(\alpha)}{\alpha+v^*}}.$$

We can write $P(t)/(t+v^*) = Q(t) + c/(t+v^*)$ for some polynomial $Q \in \mathbb{Z}_p[t]$ of degree at most $q-1$ and $c \in \mathbb{Z}_p$. Since $v^* \notin \{v_1, \dots, v_n\}$, we have $c \neq 0$. Thus,

$$\sigma^* = g_1^{Q(\alpha)+c/(\alpha+v^*)},$$

and $(v^*, (\frac{\sigma^*}{g_1^{Q(\alpha)}})^{1/c})$ is a solution to the q-SDH problem. If $c = 0 \pmod p$, then we should abort. It follows that $Pr[W_2] \geq (1 - 1/p)Pr[W_1]$.

Obviously, W_2 be the event that $\mathcal{C}$ outputs a solution to the q-SDH problem, so $Pr[W_2] = q - SDH - Adv[\mathcal{C}]$. $\square$

8 Future Work

In the future, we will continue exploring the extensibility of this scheme. Firstly, in the above scheme, if one of identity attributes is updated or removed, or a new attribute is added, the user should re-compute the full signature, and update the one stored in the access controlling server. That is very time consuming and costly, since the computation is heavy and the client side usually have limited computational resources. Thus, how to improve this scheme on updating the full signature is our future work. Secondly, how can the access control server communicate more efficiently with the cloud service and what can be done if it sends a wrong partial signature are not considered in the current presented scheme. It will be our second future work. In addition, a more sophisticated access control protocol could be studied to further reduce the trust on the semi-trusted access control server in the current scheme.

9 Conclusion

In this paper, we define a slight homomorphic signature that is suitable for access controlling in cloud computing. In this system, user's full signature on all identity attributes is stored in the access controlling server in advance, and then, if user want to apply a cloud service, which has a special requirement on one identity attribute, he only needs to secretly send the cloud service name to the access controlling server. According to the user's instruction, the access controlling server can compute a partial signature on the special identity attribute, and send it to the cloud server for identification.

We provide a secure model of this homomorphic signature, and design a scheme from Boneh–Boyen signature. We prove that our scheme is secure under q-SDH assumption with a weak adversary. Finally, we propose the future work.

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